Course Name: Geographic Information System

Week 12 - Assignment 12 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10 **Total marks: 10x1= 10**

QUESTION 1

Open-source software generally allows users to _____.

a) Freely redistribute software
b) Access source code
c) Download and use without licensing fees
d) All of the above

Answer: (d)

Detailed Solution:

Open-source software provides free access, redistribution rights, and source code availability for modification and use.

QUESTION 2

OSSIM stands for _____.

a) Open Source Software Image Map
b) Open Spatial System Image Model
c) Open Source Spatial Interface Model
d) Operational Satellite System Image Mapper

Answer: (a)

Detailed Solution:

OSSIM is an open-source software framework used for remote sensing, image processing, and geospatial applications.

QUESTION 3

GDAL is mainly used for handling _____ geospatial data formats.

a) Vector
b) Raster

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c) Tabular
d) Statistical

Answer: (b)

Detailed Solution:

GDAL (Geospatial Data Abstraction Library) supports reading and writing raster geospatial data formats.

QUESTION 4

OGR library is mainly associated with _____ data formats.

a) Raster
b) Vector
c) Image
d) Metadata

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Answer: (b)

Detailed Solution:

OGR is the vector component of GDAL used for handling vector geospatial datasets.

QUESTION 5

GRASS GIS can process which type of spatial data?

a) Raster only
b) Vector only
c) Both raster and vector
d) Neither raster nor vector

Answer: (c)

Detailed Solution:

GRASS GIS supports analysis and processing of both raster and vector spatial datasets.

QUESTION 6

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FRAGSTATS is mainly used for _____.

a) Spatial pattern analysis
b) Database encryption
c) GPS navigation
d) Image georeferencing

Answer: (a)

Detailed Solution:

FRAGSTATS analyzes landscape patterns and spatial configurations in categorical maps.

QUESTION 7

In GeoServer, WFS stands for _____.

a) Web Functional Service
b) Web Feature Service
c) Web File Service
d) Web Format Service

Answer: (b)

Detailed Solution:

WFS provides access to geographic feature data across the web using standard protocols.

QUESTION 8

OGC stands for _____.

a) Open Geospatial Committee
b) Open Graphical Consortium
c) Open Geospatial Consortium
d) Operational Geospatial Consortium

Answer: (c)

Detailed Solution:

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Open Geospatial Consortium develops standards for geospatial and location-based services.

QUESTION 9

Information describing the source, creation method, and quality of spatial data is known as _____.

a) Metadata
b) Indexing
c) Overlay
d) Classification

Answer: (a)

Detailed Solution:

Metadata provides details about spatial data such as origin, date, accuracy, and processing methods.

QUESTION 10

Which of the following are open-source operating systems?

a) Ubuntu
b) Linux
c) Red Hat
d) All of the above

Answer: (d)

Detailed Solution:

Ubuntu, Linux, and Red Hat are examples of open-source operating systems widely used in GIS and server environments.

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